MAHATMA GANDHI UNIVERSITY, KOTTAYAM

Fifth Semester Regular Examination, October 2026

MG5DSCPHY301 – Introduction to Quantum Mechanics

Scheme of Valuation

Part A – Short Answer (3 marks each) – Answer any 10

Q1. State the normalization condition for a wave function and explain its physical meaning. [U/CO1]

Answer: $\int_{-\infty}^{\infty} |\psi(x,t)|^2\, dx = 1.$ – **1.5 Marks**

The particle must be found somewhere in space with certainty, so the total probability is unity. – **1.5 Marks**

Q2. According to Born's statistical interpretation, what does $|\psi(x,t)|^2$ represent? [U/CO1]

Answer: It is the probability density. – **1.5 Marks**

$|\psi|^2 dx$ is the probability of finding the particle between x and $x+dx$ at time t – **1.5 Marks**

Q3. Define probability interference mathematically for electrons and describe how the total probability curve differs from simple classical addition. [U/CO1]

Answer: With amplitudes ψ_1, ψ_2 from the two slits,

$$P_{12} = |\psi_1 + \psi_2|^2 = P_1 + P_2 + 2\sqrt{P_1 P_2}\cos\delta,$$

δ being the phase difference. The cross term produces alternating maxima and minima (zero), so the curve is not the sum of the single-slit curves. – **1.5 Marks**

Classically $P_{12} = P_1 + P_2$ (smooth curve). – **1.5 Marks**

(Give credits even if only curves are drawn)

Q4. What is an "ideal experiment" according to Richard Feynman? Give an example of an event. [U/CO1]

Answer: 'An ideal experiment is one in which all of the initial and final conditions of the experiment are completely specified.' E.g.-: an electron arriving at the detector and producing a click at a particular position x. – **3 Marks**

Q5. For a particle with energy $E > V_0$ approaching a step potential, write the expressions for the wave numbers k (incident region) and l (transmitted region). [A/CO2]

Answer: $k = \dfrac{\sqrt{2mE}}{\hbar}, \qquad l = \dfrac{\sqrt{2m(E-V_0)}}{\hbar}.$ – **3 Marks**

Q6. Define the operators a_+ and a_-. Why is the ordering of these operators important? [U/CO2]

Answer: $a_\pm = \dfrac{1}{\sqrt{2\hbar m\omega}}\,(m\omega\hat{x} \mp i\hat{p}).$ – **2 Marks**

They do not commute: $[a_-, a_+] = 1$. Hence $a_+a_- \neq a_-a_+$ and the Hamiltonian takes different forms: $H = \hbar\omega(a_+a_- + \frac{1}{2}) = \hbar\omega(a_-a_+ - \frac{1}{2})$. – **1 Marks**

Q7. Determine whether the vectors $(1, 2, 0), (0, 1, 1), (1, 3, 1)$ are linearly independent. Justify. [A/CO3]

Answer: Let $v_1 = (1, 2, 0)$, $v_2 = (0, 1, 1)$, $v_3 = (1, 3, 1)$. We can see that $v_1 + v_2 = v_3$. (OR The determinant is $1(1 - 3) - 2(0 - 1) + 0 = -2 + 2 = 0$.) So the vectors are linearly dependent. – **3 Marks**

Q8. Show that $\sin x$ and $\sin 2x$ are orthogonal on $[-\pi, \pi]$. [A/CO3]

Answer: $\displaystyle\int_{-\pi}^{\pi} \sin x \sin 2x \, dx = \tfrac{1}{2}\int_{-\pi}^{\pi} (\cos x - \cos 3x) \, dx = \tfrac{1}{2}\Big[\sin x - \tfrac{\sin 3x}{3}\Big]_{-\pi}^{\pi} = 0.$ – **3 Marks**

Q9. Write down the three properties of eigenvalues and eigenvectors under a Hermitian transformation. [U/CO3]

Answer: (i) Eigenvalues are real. (ii) Eigenvectors of distinct eigenvalues are orthogonal. (iii) The eigenvectors form a complete set, so the matrix can be diagonalised by a unitary transformation. – **3 Marks**

Q10. Calculate the minimum uncertainty in velocity of an electron confined within a region of width 1 Å. [A/CO4]

Answer: $\Delta x \Delta p \geq \dfrac{\hbar}{2}$ or $\Delta x \, (m\Delta v) \geq \dfrac{\hbar}{2}$ – **1 Marks**

For minimum uncertainty, equality holds;

$$\Delta v = \frac{\hbar}{2m\Delta x} = \frac{1.055 \times 10^{-34}}{2(9.11 \times 10^{-31})(10^{-10})} \approx 5.8 \times 10^5 \text{ m/s}.$$ – **2 Marks**

Q11. Why must a wave function be normalized? [U/CO4]

Answer: $|\psi|^2$ is probability density. The total probability of finding the particle must be 1. Only a normalized wave function gives meaningful probabilities. – **3 Marks**

Q12. Show that the eigenfunctions of the position operator are not in Hilbert space. [U/CO4]

Answer:

$\hat{x}\, g(x) = x_0 g(x)$
$\Rightarrow (x - x_0)g = 0$
$\Rightarrow g(x) = A\, \delta(x - x_0).$ – **1 Marks**

Then $\int |g|^2 dx = |A|^2 \delta(0) = \infty$, so g is not square-integrable and lies outside Hilbert space. – **2 Marks**

Part B – Short Essay (8 marks each) – Answer any 5

Q13. Explain how Newton's second law in classical mechanics is analogous to the Schrödinger equation in quantum mechanics in explaining the time evolution of a physical system. [U/CO1]

Answer:

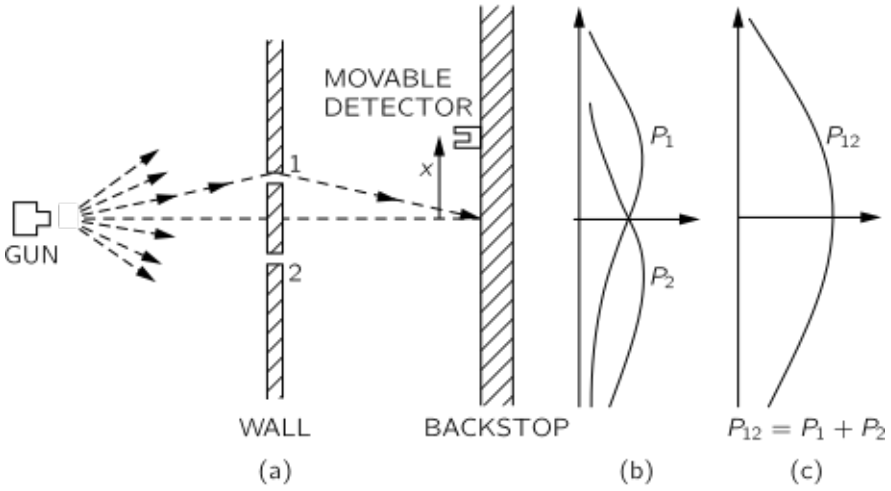
	Classical	Quantum
Equation	$F = m\,d^2x/dt^2$	$i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$
State	(x, v)	$\Psi(x,t)$
Initial data	$x(0), v(0)$	$\Psi(x,0)$
Output	trajectory $x(t)$	$\Psi(x,t)$, probabilities $\lvert\Psi\rvert^2$

- Both are deterministic: the initial state fixes the state at all later times.
- Newton's law is second order in time; Schrödinger's is first order.
- In Newton's law the force is $F = -dV/dx$; in quantum mechanics V enters through $\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V$.
- They differ in what is predicted: exact path versus probabilities of outcomes.
- Ehrenfest's theorem links them: $d\langle p\rangle/dt = -\langle dV/dx\rangle$.

– Any 4 points 8 Marks

Q14. Describe the double-slit experiment for classical particles (bullets) and waves. With neat diagrams, explain the distribution patterns for single-slit and double-slit conditions. [U/CO1]

Answer: Bullets. They arrive in lumps. With one slit open the distributions are P_1, P_2 (single hump behind each slit). With both open, $P_{12} = P_1 + P_2$: no interference.



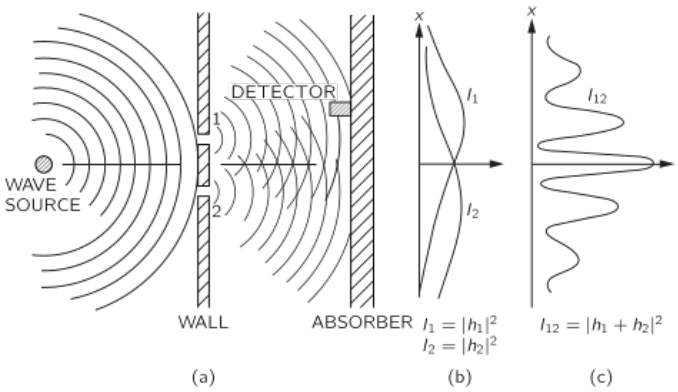
– 4 Marks

Waves. Intensity is the square of the amplitude. With one slit open: broad diffraction humps I_1, I_2. With both open,

$$I_{12} = |h_1 + h_2|^2 = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta, \quad \delta = \frac{2\pi d \sin\theta}{\lambda}.$$

Maxima: $d\sin\theta = n\lambda$; minima: $d\sin\theta = (n + \frac{1}{2})\lambda$, giving bright and dark fringes. Electrons arrive as lumps like bullets but build up the wave pattern.

Waves: $I_{12} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta$ *(fringes within the diffraction envelope):*



– **4 Marks**

Q15. Evaluate the normalization constant for the wave function $\psi = A\sin(2\pi x/a)$. [A/CO2]

Answer: Assume one-dimensional infinite square well of width a (not given in the question). Taking the region $0 \le x \le a$:

$$|A|^2 \int_0^a \sin^2 \frac{2\pi x}{a} dx = |A|^2 \frac{a}{2} = 1 \;\Rightarrow\; A = \sqrt{\frac{2}{a}}.$$

– **8 Marks**

Give full credits, if the interval is assumed symmetric from $-a$ to $+a$. Then, the integral yields a, giving $A = \frac{1}{\sqrt{a}}$ or for any other interval and for arriving at correct expression / normalization constant – **8 Marks**

Q16. Solve the time-independent Schrödinger equation for a free particle in one dimension. Write the general solution and discuss the significance of the energy eigenvalues. [U/CO2]

Answer: With $V = 0$: $-\frac{\hbar^2}{2m}\psi'' = E\psi \quad \Rightarrow \psi'' + k^2\psi = 0$, $k = \pm\sqrt{2mE}/\hbar$.

General solution: $\psi(x) = Ae^{ikx} + Be^{-ikx}$; with time dependence, $\Psi = Ae^{i(kx-\omega t)} + Be^{-i(kx+\omega t)}$, $\omega = E/\hbar$. – **4 Marks**

Significance:

- $E = \hbar^2k^2/2m$ is purely kinetic; every $E \ge 0$ is allowed, so the spectrum is continuous (no quantization).
- Each E is doubly degenerate ($\pm k$): right- and left-moving waves.
- These solutions are not normalizable ($|\psi|^2$ constant); a physical particle is a wave packet $\Psi = \int \phi(k)e^{i(kx-\hbar k^2 t/2m)}dk$.
- The packet moves with group velocity $\hbar k/m$, equal to the classical speed and twice the phase velocity. – **4 Marks**

Q17. Define the Hamiltonian operator. Derive the time-independent Schrödinger equation in terms of the Hamiltonian operator. [U/CO2]

Answer: The Hamiltonian is the total-energy operator:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

. – **2 Marks**

Start with $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$, V independent of t. Put $\Psi = \psi(x)f(t)$: – **2 Marks**

$$i\hbar\frac{1}{f}\frac{df}{dt} = \frac{1}{\psi}\hat{H}\psi.$$

The left side depends only on t, the right only on x, so both equal a constant E. Then $f(t) = e^{-iEt/\hbar}$ and

$$\boxed{\hat{H}\psi = E\psi} \quad \text{i.e.} \quad -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi,$$

an eigenvalue equation: E is the energy eigenvalue, ψ the eigenfunction. – **4 Marks**

Q18. Let $T = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$. (a) Find all eigenvalues of T by its characteristic equation. (b) Determine the eigenvectors for each eigenvalue. [A/CO3]

Answer:

(a) $\det(T - \lambda I) = 0$ – **1 Marks**

$(2-\lambda)\big[(2-\lambda)(1-\lambda)\big] = (2-\lambda)^2(1-\lambda) = 0$, so $\lambda = 1$ and $\lambda = 2$ (twice). – **3 Marks**

(b) $\lambda = 1$: $x + y - z = 0,\ y = 0 \qquad \Rightarrow v = (1, 0, 1)^T$.

$\lambda = 2$: $y - z = 0$, let $x = 1 \qquad \Rightarrow v = (1, 0, 0)^T$ and $(0, 1, 1)^T$.

Three independent eigenvectors exist, so T is diagonalizable. – **4 Marks**

Q19. (a) Prove that the product of two unitary matrices is unitary. Is the sum of two unitary matrices also unitary? Explain. (b) Is the product of two Hermitian matrices necessarily Hermitian? Explain. [A/CO3]

Answer: (a) If $U_1U_1^\dagger = U_2U_2^\dagger = I$ then

$$(U_1U_2)(U_1U_2)^\dagger = U_1U_2U_2^\dagger U_1^\dagger = U_1U_1^\dagger = I$$

so the product is unitary. – **2 Marks**

For the sum, $(U_1 + U_2)(U_1 + U_2)^\dagger = 2I + U_1U_2^\dagger + U_2U_1^\dagger \neq I$
e.g. $I + I = 2I$ gives $(2I)(2I)^\dagger = 4I \neq I$. The sum is NOT unitary in general. – **2 Marks**

(b) $(AB)^\dagger = B^\dagger A^\dagger = BA$, so AB is Hermitian only if $AB = BA$. Hence the product of two Hermitian matrices not necessarily be a Hermitian matrix;
e.g. $\sigma_x\sigma_z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is not Hermitian. – **4 Marks**

Q20. Show that $e^{-x^2/2}$ is an eigenfunction of the operator $\hat{Q} = \dfrac{d^2}{dx^2} - x^2$ and determine the eigenvalue. [A/CO4]

Answer: Let $\psi = e^{-x^2/2}$. Then $\psi' = -x\psi, \qquad \psi'' = -\psi + x^2\psi = (x^2 - 1)\psi$.

So

$$\hat{Q}\psi = \psi'' - x^2\psi = (x^2 - 1 - x^2)\psi = -\psi.$$

Hence ψ is an eigenfunction with eigenvalue -1. – **8 Marks**